

DATE : \_\_\_\_\_

Q. In a Latin Square design with  $p$  treatments,  $p$  rows and  $p$  columns, the observation is assumed to be represented by the model,

$$Y_{ijk} = \mu + \alpha_i + \delta_j + \gamma_k + \beta(X_{ijk} - \bar{X}) + E_{ijk}; i, j, k = 1, 2, \dots, p$$

Obtain the least square estimates of the parameters of the above model.

$$Y_{ijk} = \mu + \alpha_i + \delta_j + \gamma_k + \beta(X_{ijk} - \bar{X}) + E_{ijk}$$

$Y_{ijk}$ : obs. values

$X_{ijk}$ : Covariate values

$\mu$ : Overall mean

$\alpha_i, \delta_j, \gamma_k$ : Row, column & treatment effect respectively.

$\beta$ : regression coefficient

$$E_{ijk} = \text{RSS}$$

So, we need to minimize the Residual sum of squares (RSS),

$$\text{RSS} = \sum_{i,j,k}^p [Y_{ijk} - (\mu + \alpha_i + \delta_j + \gamma_k + \beta(X_{ijk} - \bar{X}))]^2 \quad \text{--- (A)}$$

To estimate the parameters, we take Partial

DATE : \_\_\_\_\_

derivatives of RSS w.r.t.  $\mu, \alpha_i, \beta, \delta_j, \gamma_k$  and set them equal to zero.

$\Rightarrow$  Partial derivative of (A) w.r.t. " $\mu$ "

$$\frac{\partial \text{RSS}}{\partial \mu} = 2 \sum_{ijk}^P [y_{ijk} - (\mu + \alpha_i + \delta_j + \gamma_k + \beta(x_{ijk} - \bar{x}))] (-1)$$

$$= -2 \sum_{ijk}^P [y_{ijk} - (\mu + \alpha_i + \delta_j + \gamma_k + \beta(x_{ijk} - \bar{x}))]$$

Equating it to zero

$$\sum_{ijk}^P [y_{ijk} - (\mu + \alpha_i + \delta_j + \gamma_k + \beta(x_{ijk} - \bar{x}))] = 0$$

$$\sum_{ijk}^P y_{ijk} = \sum_{ijk}^P \mu + \sum_{ijk}^P \alpha_i + \sum_{ijk}^P \delta_j + \sum_{ijk}^P \gamma_k + \beta \sum_{ijk}^P (x_{ijk} - \bar{x})$$

Here,

' $\mu$ ' is constant doesn't vary with  $i, j, k$ .

$$\sum_{ijk}^P y_{ijk} = P^2 \cdot \mu + \sum_{ijk}^P \alpha_i + \sum_{ijk}^P \delta_j + \sum_{ijk}^P \gamma_k + \beta \sum_{ijk}^P (x_{ijk} - \bar{x})$$

We assume sum-to-zero constraints on the row, column and treatment effects.

$$\sum_i \alpha_i = 0, \sum_j \delta_j = 0, \sum_k \gamma_k = 0$$

therefore

$$\sum_{ijk} \alpha_i = \sum_{ijk} \delta_j = \sum_{ijk} \gamma_k = 0$$

Then

DATE : \_\_\_\_\_

$$\sum_{ijk}^p x_{ijk} = p^2 \cdot \mu + \beta \sum_{ijk}^p (x_{ijk} - \bar{x})$$

Sum of deviation taken from  $\sum_{ijk}^p (x_{ijk} - \bar{x})$  is zero

Thus

$$\sum_{ijk}^p x_{ijk} = p^2 \cdot \mu$$

$$\mu = \frac{\sum_{ijk}^p x_{ijk}}{p^2}$$

"note"

The appearance of " $p^2$ " comes from Latin Square design assumes there are  $p$  rows,  $p$  columns and  $p$  treatments, resulting in  $p^2$  total obs.

Hence

$$\mu = \frac{\sum_{ijk}^p x_{ijk}}{p^2} \quad \text{or} \quad \frac{\text{sum of all obs } (y_{ijk})}{\text{Total no. of obs } (p^2)}$$

$\Rightarrow$  Partial derivative of (A) w.r.t " $\alpha_i$ "

$$\frac{\partial \text{RSS}}{\partial \alpha_i} = 2 \sum_{jk} [x_{ijk} - (\mu + \alpha_i + \delta_j + \delta_k + \beta(x_{ijk} - \bar{x}))] \quad (-1)$$

$$= 2 \sum_{jk} [x_{ijk} - (\mu + \alpha_i + \delta_j + \delta_k + \beta(x_{ijk} - \bar{x}))]$$

Equating it to zero

DATE : \_\_\_\_\_

$$\sum_{jk} [Y_{ijk} - (\mu + \alpha_i + \delta_j + \gamma_k + \beta(X_{ijk} - \bar{X}))] = 0$$

$$\sum_{jk} Y_{ijk} = \sum_{jk} \mu + \sum_{jk} \alpha_i + \sum_{jk} \delta_j + \sum_{jk} \gamma_k + \beta \sum_{jk} (X_{ijk} - \bar{X})$$

$$\sum_{jk} Y_{ijk} = P \cdot \mu + P \alpha_i \quad (\because \text{due to row effect})$$

$$= P(\mu + \alpha_i)$$

$$\frac{\sum_{jk} Y_{ijk}}{P} = \mu + \alpha_i$$

$$\frac{\sum_{jk} Y_{ijk}}{P} - \mu = \alpha_i$$

$$\alpha_i = \frac{\sum_{jk} Y_{ijk}}{P} - \mu$$

$\Rightarrow$  Partial derivative of (A) w.r.t. " $\delta_j$ "

$$\frac{\partial RSS}{\partial \delta_j} = 2 \sum_{ik} [Y_{ijk} - (\mu + \alpha_i + \delta_j + \gamma_k + \beta(X_{ijk} - \bar{X}))] (-1)$$

$$= -2 \sum_{ik} [Y_{ijk} - (\mu + \alpha_i + \delta_j + \gamma_k + \beta(X_{ijk} - \bar{X}))]$$

Equating it to zero

$$\sum_{ik} [Y_{ijk} - (\mu + \alpha_i + \delta_j + \gamma_k + \beta(X_{ijk} - \bar{X}))] = 0$$

$$\sum_{ik} Y_{ijk} = \sum_{ik} \mu + \sum_{ik} \alpha_i + \sum_{ik} \delta_j + \sum_{ik} \gamma_k + \beta \sum_{ik} (X_{ijk} - \bar{X})$$

$$\sum_{ik} Y_{ijk} = P \cdot \mu + P \delta_j \quad (\because \text{due to column effect})$$

DATE : \_\_\_\_\_

$$= P(\mu + \delta_j)$$

$$\mu + \delta_j = \frac{\sum_{ik} Y_{ijk}}{p}$$

$$\delta_j = \frac{\sum_{ik} Y_{ijk}}{p} - \mu$$

$\Rightarrow$  Partial derivative of (A) w.r.t. " $\delta_k$ "

$$\frac{\partial \text{RSS}}{\partial \delta_k} = 2 \sum_{ij} [Y_{ijk} - (\mu + \alpha_i + \delta_j + \delta_k + \beta(X_{ijk} - \bar{X}))](-1)$$

$$= -2 \sum_{ij} [Y_{ijk} - (\mu + \alpha_i + \delta_j + \delta_k + \beta(X_{ijk} - \bar{X}))]$$

Equating it to zero

$$\sum_{ij} [Y_{ijk} - (\mu + \alpha_i + \delta_j + \delta_k + \beta(X_{ijk} - \bar{X}))] = 0$$

$$\sum_{ij} Y_{ijk} = \sum_{ij} \mu + \sum_{ij} \alpha_i + \sum_{ij} \delta_j + \sum_{ij} \delta_k + \beta \sum_{ij} (X_{ijk} - \bar{X})$$

$$\sum_{ij} Y_{ijk} = P \cdot \mu + P \cdot \delta_k \quad (\because \text{due to treatment effect})$$
$$= P(\mu + \delta_k)$$

$$\mu + \delta_k = \frac{\sum_{ij} Y_{ijk}}{p}$$

$$\delta_k = \frac{\sum_{ij} Y_{ijk}}{p} - \mu$$

DATE: \_\_\_\_\_  
⇒ Partial derivative of (A) w.r.t. "β"

$$\frac{\partial RSS}{\partial \beta} = 2 \sum_{ijk} [Y_{ijk} - (\mu + \alpha_i + \delta_j + \gamma_k + \beta(X_{ijk} - \bar{X}))] (-1) (X_{ijk} - \bar{X})$$

Equating to zero

$$\left[ \sum_{ijk} Y_{ijk} - \sum_{ijk} \mu - \sum_{ijk} \alpha_i - \sum_{ijk} \delta_j - \sum_{ijk} \gamma_k - \beta \sum_{ijk} (X_{ijk} - \bar{X}) \right] (X_{ijk} - \bar{X}) = 0$$

$$\left[ \sum_{ijk} [Y_{ijk} - (\mu + \delta_j + \alpha_i)] - \beta \sum_{ijk} (X_{ijk} - \bar{X}) \right] (X_{ijk} - \bar{X}) = 0$$

$$\sum_{ijk} (Y_{ijk} - \mu - \alpha_i - \delta_j - \gamma_k) (X_{ijk} - \bar{X}) - \beta \sum_{ijk} (X_{ijk} - \bar{X})^2 = 0$$

$$\beta \sum_{ijk} (X_{ijk} - \bar{X})^2 = \sum_{ijk} (Y_{ijk} - \mu - \alpha_i - \delta_j - \gamma_k) (X_{ijk} - \bar{X})$$

$$\beta = \frac{\sum_{ijk} (Y_{ijk} - \mu - \alpha_i - \delta_j - \gamma_k) (X_{ijk} - \bar{X})}{\sum_{ijk} (X_{ijk} - \bar{X})^2}$$